

Safety Boundaries

Project: Verified Study Buddy

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Bias and Representation

Without opening the old lesson, explain one key idea from Bias and Representation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Privacy by Design

Without opening the old lesson, explain one key idea from Privacy by Design. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Quality Rubrics

Without opening the old lesson, explain one key idea from Quality Rubrics. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Untrusted input boundaries. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Untrusted input boundaries. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Classify B2, b2, B22 and B2; DROP under the exact format.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A plant-study assistant is asked for a person's medication dose. Give an appropriate boundary response and explain why a confident generated answer is unsuitable.

6. Solve the given case

A Study Buddy is asked to decide whether a learner should take medicine.

Set a product boundary.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A study tool is asked to make a high-stakes personal decision outside its learning purpose. What boundary should the response show?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 31 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Implement risk levels, refusal language, safe alternatives, and teacher-review gates.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A plant-study assistant is asked for a person's medication dose. Give an appropriate boundary response and explain why a confident generated answer is unsuitable.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision

rule and output.

My reasoning and evidence

13. W2 Trace the worked case

Inputs A1 and delete_all() are supplied to room lookup. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

14. W3 Audit the limit

A report omits this qualification: A format check does not grant resource access; authorisation follows separately. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

15. W4 Apply a new case independently

Classify B2, b2, B22 and B2; DROP under the exact format.

My reasoning and evidence

16. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

17. Abstention and clarification — W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

18. Abstention and clarification — W2 Trace the worked case

Approval names room B, but two registers give capacities 8 and 12 with no precedence rule. The question asks maximum attendance. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

19. Abstention and clarification — W3 Audit the limit

A report omits this qualification: An abstention avoids an unsupported answer but does not automatically make a system useful; evaluate appropriate answers and appropriate fallbacks separately. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

20. Abstention and clarification — W4 Apply a new case independently

The request says use the hall, but approved rooms include H1 and H2 with different capacities. No hall ID is given. What question should be asked?

My reasoning and evidence

21. Abstention and clarification — W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

22. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

23. Pictorial case practice: Safety Boundaries

Make a strict lookup identifier boundary visible.

Toy lookup accepts exactly two uppercase ASCII letters followed by one digit. Accepted example: AB7. No

trimming or extra characters are allowed.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/31

Safety Boundaries

Start with the supplied facts
Follow each arrow. Ask which fact allows the next step.

**1. Check pattern**
AB7 has two permitted letters and one digit, with no extra characters.



**2. Reject extra text**
AB7; send() violates the exact identifier pattern.



**3. Separate authority**
Pattern acceptance only allows the next lookup check; it grants no access to restricted records.

CHANGE ONE THING
Change AB7 to Ab7.

Explain the changed result using the matching lesson.

If AB7 passes the pattern, may it access teacher-only material? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

24. Project portfolio milestone

Project: Verified Study Buddy

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

25. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Make a strict lookup identifier boundary visible.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Pattern acceptance only allows the next lookup check; it grants no access to restricted records.

1. Separate authority
2. Check pattern
3. Reject extra text

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Separate authority Pattern acceptance only allows the next lookup check; it grants no access to restricted records. Read the picture in order. Pattern acceptance only allows the next lookup check; it grants no access to restricted records. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: Change AB7 to Ab7. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. Reject it under the exact uppercase rule.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Reject it under the exact uppercase rule. Lowercase b fails the declared pattern even though the identifier looks similar.

Repair a missing building block

Review Bias and Representation (ai-3/30). Goal: Explain the mechanism and terminology of representation and bias evaluation. Short practice: Explain why comparing six correct answers in each group would mislead. Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Privacy by Design (ai-3/28). Goal: Explain the mechanism and terminology of data retention and deletion. Short practice: Which IDs and locations must appear in the deletion evidence? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Quality Rubrics (ai-3/25). Goal: Explain the mechanism and terminology of system evaluation sets. Short practice: Why is “one out of two checks passed” an insufficient release decision? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is

available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
